1)Table: Player

- 1NF: All fields (e.g., nickname, king_level) contain atomic values. There are no repeating groups like card_1, card_2 or comma-separated lists
- 2NF: The PK (player_id) is a single attribute. Partial dependency is impossible by definition
- 3NF: All non-key attributes depend solely on player_id

2)Table: Clan

- 1NF: Attributes like location and clan_tag are atomic
- 2NF: PK is a single attribute. No partial dependencies
- 3NF: Satisfied. While 'members_count' might appear to be derived data (suggesting denormalization), this database stores only a subset of all players. Since not all clan members exist in the 'Player' table, 'members_count' cannot be calculated via a simple COUNT() query. Therefore, it is a non-redundant, atomic attribute essential for data integrity.

3)Table: Clan_War

- 1NF: Values are atomic
- 2NF: PK is single attribute
- 3NF: result and place depend on the specific war instance (war_id), not the clan_id. The schema correctly links to Clan via ID only

4)Table: Deck

- 1NF, 2NF, 3NF**:** This is a simple dictionary/lookup table. It creates a distinct entity for a Deck, referenced by players and matches, avoiding redundancy in other tables.

5)Table: Deck_Card

- 1NF: Atomic IDs
- 2NF: There are no non-key attributes to check for partial dependencies. The table strictly defines the Many-to-Many relationship between Decks and Cards
- 3NF: No non-key attributes, so no transitive dependencies exist

6)Table: Card

- 1NF: Atomic
- 2NF: Single PK
- 3NF: Attributes like elixir_cost describes the card itself. If we stored deck_win_rate here, it would violate 3NF, but the current schema is clean

7)Table: Tournament

- 1NF/2NF: Standard single-key entity
- 3NF: location depends on the tournament instance. No transitive dependencies found

8)Table: Match

- 1NF: Atomic

- 2NF: Single PK
- 3NF: Denormalized. 'winner_id' is technically derived from Player_Stats but is stored here to allow fast history querying without joins

9)Table: Player_Stats

- 1NF: Atomic
- 2NF:
    - Does total_damage depend only on player_id? No (that would be lifetime stats)
    - Does it depend only on match_id? No (there are two players in a match)
    - It depends on the combination of Player+Match. Thus, 2NF is achieved
- 3NF: No transitive dependencies. The deck_id is the deck used specifically in this context

10)Table: Player_Penalty

- 1NF/2NF/3NF: Standard entity tracking events. rule_code implies there might be a separate Rules table in a stricter schema, but assuming rule_code is just a string identifier here, it satisfies 3NF

11)Table: Player_Tournament

- 1NF: Atomic
- 2NF: final_rank depends on the specific player's performance in that specific tournament. It cannot be determined by player_id alone or tournament_id alone
- 3NF: No transitive dependencies